

Satellite Networks for Continuous Zonal Coverage

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Solutions are obtained to the problem of providing continuous surveillance of latitudinally bounded zones of the globe by means of a particular family of satellite networks. Each network of the family considered consists of satellites equally divided among, and uniformly distributed around, circular orbits of common altitude; the orbit planes are equally inclined with respect to the Equator and symmetrically arranged about the polar axis. Relations are obtained between the number and distribution of satellites used, the characteristics of the network in which they are arranged, and the extent of the continuous zonal coverage they accomplish. These relations are then employed to select from the family of networks considered those which minimize the number of satellites required as a function of altitude for a variety of coverage problems. Results obtained for several representative cases are given graphically.

IT MAY well be desirable in the near future to establish orbital systems which afford continuous coverage of either all or a limited portion of the globe. A large variety of missions requiring such systems are described in the literature (1,2,3).² An important consideration in the feasibility of such systems will be the number of satellites and the orbital altitudes required to accomplish the desired coverage missions. Recently Vargo (4) presented a solution to the problem of obtaining uninterrupted coverage of the entire globe by means of a particular network configuration. The present analysis treats a family of networks which provide continuous coverage of latitudinally bounded zones on Earth. Using relations obtained between network characteristics and the resultant coverage, one can select those networks which require the fewest satellites as a function of altitude for a variety of coverage problems.

Assumptions

The networks considered in this analysis consist of satellites equally divided among, and uniformly distributed around, circular orbits of common altitude; the orbit planes are equally inclined with respect to the Equator and are symmetrically arranged about the polar axis. Each satellite carries a sensor which observes the segment of the globe subtended by the scanned solid cone whose axis is continually aligned with the local vertical. The globe is assumed spherical, and no differential perturbations are assumed to act (i.e., no perturbations which would alter the *relative* orientation of satellites in any orbit plane or the angular spacing between orbit planes). Results obtained by King-Hele (5) show that for identically inclined, equal circular orbits, perturbations resulting from Earth oblateness will not, in fact, produce a change in the interorbit spacing; one can compensate for the radial excursions executed by the satellites (and the non-spherical shape of Earth) by overdesigning the network.

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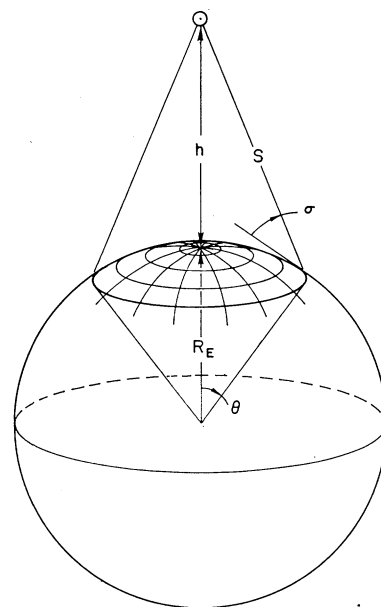
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² Numbers in parentheses indicate References at end of paper.

Analytical Development

General Principles

A satellite at altitude h can view a spherical segment whose size θ (measured by the great circle arc from the satellite sub-point to the small circle bounding the observed segment) depends upon σ the minimum angle of visibility consistent with the sensor capability and mission objectives. Fig. 1 illustrates



- R_E RADIUS OF EARTH
- h SATELLITE ALTITUDE
- θ OBSERVED HALF-ARC
- σ MINIMUM ANGLE OF VISIBILITY
- S MAXIMUM SLANT RANGE

Fig. 1 Area observed by a single satellite

the instantaneous coverage afforded by a single satellite. From the figure and the law of sines, it follows that

$$h = R_E [\cos \sigma / \cos (\theta + \sigma) - 1] \quad [1]$$

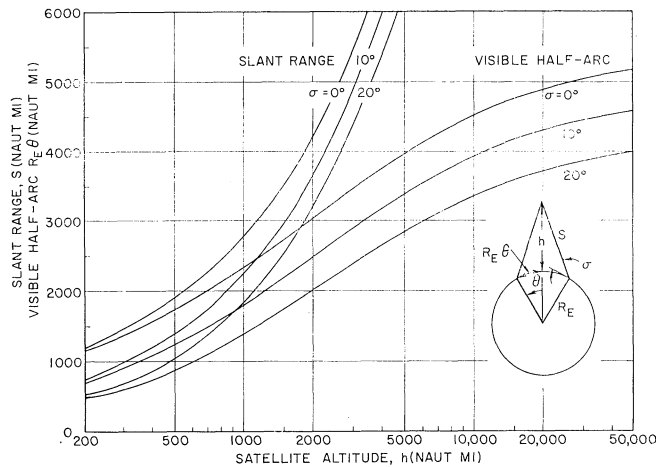


Fig. 2 Slant range and visible half-arc vs. satellite altitude and minimum viewing angle

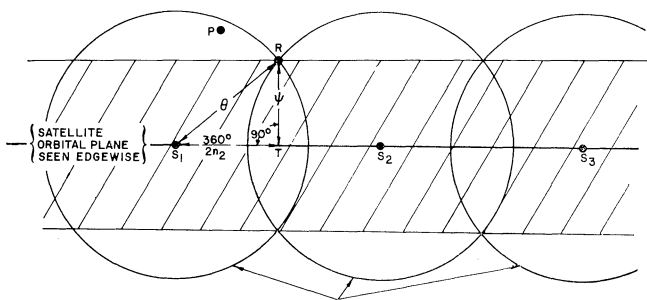


Fig. 3 Spherical segments visible from S_1 , S_2 and S_3 , respectively

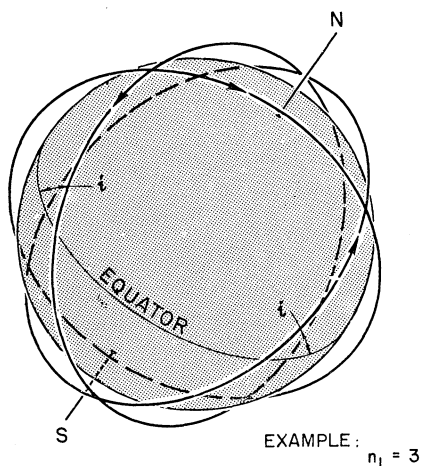


Fig. 4 Typical satellite network

NETWORK CHARACTERISTICS:

1. THE n_1 ORBITS ARE CIRCLES OF COMMON SIZE.
2. n_2 SATELLITES (NOT SHOWN HERE) ARE EVENLY SPACED IN EACH ORBIT.
3. THE ORBIT PLANES ARE EQUALLY INCLINED TO THE EQUATOR.
4. THE NODAL AXES OF THE n_1 ORBIT PLANES ARE UNIFORMLY SPACED AROUND THE EQUATOR.

where R_E is the radius of the Earth or other central body. Fig. 2 shows the relation between slant range S , observed half-arc θ , satellite altitude h and minimum viewing angle σ .

The spherical segments observed by the satellites in a common orbit plane will combine to observe a swath of angular half-width ψ on the globe. Fig. 3 illustrates the relation between great circle arcs θ and ψ and n_2 the number of satellites in each orbit plane. Although a point such as P will be uncovered periodically as the satellites advance in orbit, the shaded swath will *always* be observed from the orbit plane. By spherical trigonometry applied to triangle RS_1T

$$\cos \theta = \cos \psi \cos \pi/n_2 \quad [2]$$

The angle ψ will be found to depend upon n_1 , the number of orbit planes used, their inclination, and the location and extent of the area to be continuously observed.

Since N denotes the total number of satellites used in a network, it follows by definition that

$$N = n_1 n_2 \quad [3]$$

where n_1 and n_2 are integers.

The general network consists of n_1 orbit planes which are equally inclined with respect to the Equator and whose nodal axes are uniformly spaced around the equatorial circle. Fig. 4 shows a typical satellite network of the type treated here, and Fig. 5 gives views of such inclined systems seen from above the pole. Note that as the orbit inclination with respect to the equator tends to 90 deg, the inclined network becomes polar. If n_1 is odd, the resulting polar network will consist of n_1 distinct orbit planes. However, if n_1 is even, the polar network obtained will consist of only $n_1/2$ distinct planes; each "double plane" will contain two chains of n_2 satellites moving in opposition. This fact will be important later in the discussion.

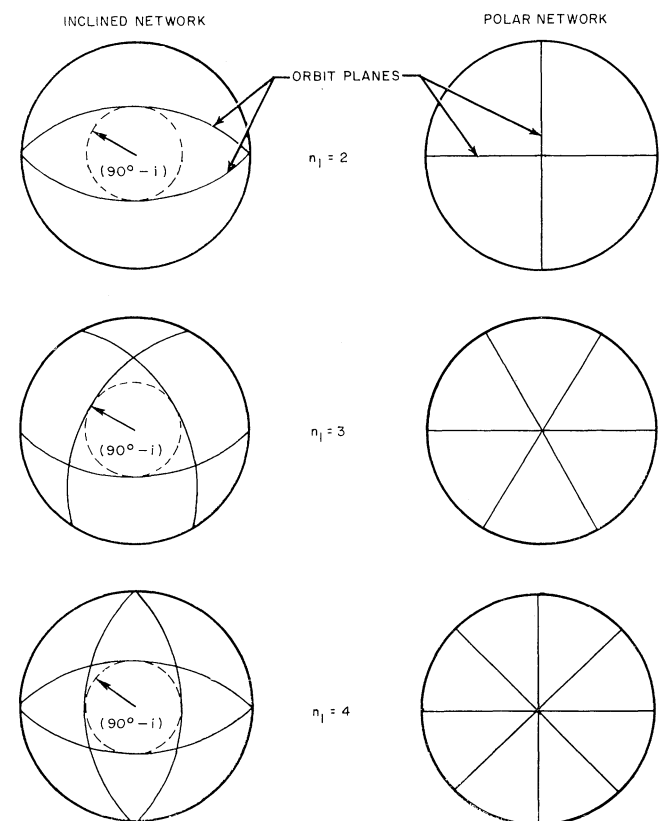


Fig. 5 Comparison of satellite networks viewed from above the pole

Method

Fig. 6 indicates the manner in which the swaths observed by satellites in adjacent orbit planes will merge to provide continuous coverage of a latitude belt in each hemisphere.

The zones viewed from the northeastbound satellites in orbit planes 1 and 2 will intersect at point S which will thus be the southernmost³ point viewed simultaneously from those planes. The southeastbound satellites in orbit planes $(n_1 - p)$ and $[n_1 - (p + 1)]$ will view swaths which intersect at point T . (This "plane index" p is chosen as the least integer such that T falls to the west of S .) Let point V at latitude λ be the southernmost point on the rotating Earth, continuously observed by the satellite network. Then V will be determined by the intersection of the northern boundaries of the unobserved areas terminating at points S and T ; by symmetry, V will fall directly below point B , the intersection of planes 1 and $(n_1 - p)$. The northernmost point N continuously observed by the network will be determined by the intersection of the northern boundaries of the swaths viewed from planes 1 and 2. Its latitude will be denoted by μ .

Combining Equations [1, 2 and 3] to eliminate θ , and applying spherical trigonometric relations to the geometry in Fig. 6 yields the following system of relations

$$h = R_E(\cos \sigma / \cos \{\arccos [\cos (\pi n_1 / N) \cos \psi] + \sigma\} - 1) \quad [4]$$

$$\lambda = \arctan \left\{ \tan i \cos \left[\left(\frac{p+1}{n_1} \right) \pi \right] \right\} - \arcsin \left(\frac{\sin \psi}{\sqrt{1 - \sin^2 i \sin^2 \{[(p+1)/n_1]\pi\}}} \right) \quad [5]$$

where

$$\psi = \arcsin [\sin \mu \cos i - \cos (\pi/n_1) \cos \mu \sin i] \quad [6]$$

and where p is the least integer such that

$$p \geq \frac{n_1}{2} - 2 - \frac{n_1}{\pi} \arctan \left[\frac{\cos i}{\sin \psi} \sqrt{\frac{\cos^2 \psi}{1 - \sin^2 i \sin^2 (\pi/n_1)} - 1} \right] \quad [7]$$

In the foregoing relations, $0 \text{ deg} < \psi < 90 \text{ deg}$. The extremes of required coverage will be specified in the range $0 \text{ deg} \leq \lambda < \mu \leq 90 \text{ deg}$, but a resulting negative value of λ or a value of μ in excess of 90 deg will indicate, respectively, coverage south of the Equator by satellites in the northern hemisphere or coverage beyond the poles. (This will be mentioned again later in the discussion.) Note that in relation [7], in order that the quantity under the radical be positive ψ must not exceed $\arccos \sqrt{1 - \sin^2 i \sin^2 (\pi/n_1)}$; this insures that the swaths observed from adjacent ascending orbits will have a discrete intersection. If ψ exceeds that value, the swaths will overlap everywhere (indeed, providing coverage to $\lambda = 0 \text{ deg}$, but making S an undefined point).

Relations [4-7] may be combined in principle to eliminate p and ψ ; then relations of the following form result

$$f_1(n_1, i, \lambda, \mu) = 0 \quad [8]$$

$$f_2(h, N/n_1, n_1, i, \mu, \sigma) = 0 \quad [9]$$

For a given coverage requirement and sensor resolution capability, the quantities λ , μ and σ will be known. Then from Equation [8] one may obtain values of i satisfying the coverage requirements for a series of integral values of n_1 . These (n_1, i) pairs when substituted in Equation [9] yield curves of N vs. h for constant n_1 and i . The envelope to the n_1 family of curves thus obtained is the locus of the optimum

³ Future remarks, applicable to either hemisphere, will be made in terms of only the northern hemisphere.

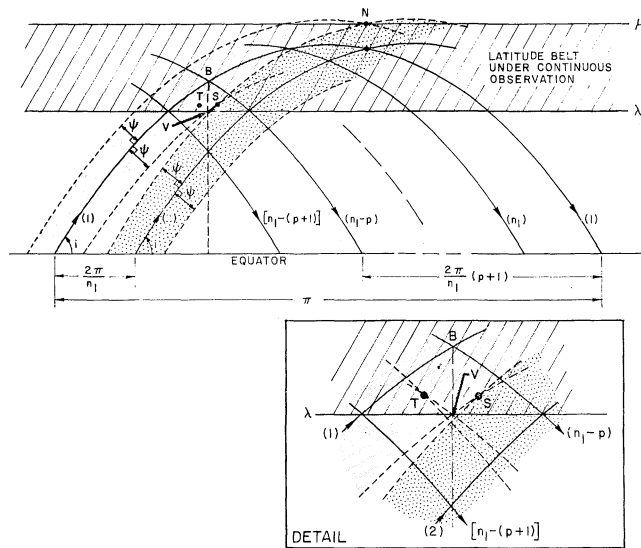


Fig. 6 Mercator projection of general coverage geometry

networks. An "optimum" network is defined here as one which accomplishes the specified coverage mission utilizing the minimum number of satellites at a given system altitude. Strictly speaking the envelope obtained is the theoretical optimum curve since both n_1 and n_2 will not, in general, assume integral values along it. The theoretical optimum curve will be useful primarily as a standard for comparison of practical networks (i.e., ones consisting of whole numbers of orbit planes containing whole numbers of satellites).

Applications

The specified coverage will always be such that $0 \text{ deg} \leq \lambda < \mu \leq 90 \text{ deg}$ and $0 \text{ deg} \leq i \leq 90 \text{ deg}$. However, if particular values are assigned to one or more of these quantities, certain special networks may be examined. Two of these, the polar and equatorial networks, are discussed in some detail in the following. The third case is concerned with the more common network where $0 \text{ deg} < i < 90 \text{ deg}$.

Case A: The Polar Network

For an orbit inclination of 90 deg , the network is polar. (The particular case $n_1 = 4$, $n_2 = 3$ is shown in Fig. 7.) Relations [5-7] then simplify to the following forms

$$p \geq (n_1/2 - 2) \quad [10]$$

$$\psi = \arcsin (\cos \lambda \cos \{[(p+1)/n_1]\pi\}) \quad [11]$$

$$\mu = \arccos [-\sin \psi / \cos (\pi/n_1)] \quad [12]$$

Equation [12] indicates that μ will exceed 90 deg for $\psi > 0 \text{ deg}$. The quantity p in inequality [10] may best be examined by considering its behavior for even and odd n_1 values. For odd n_1 , p is equal to $(n_1 - 3)/2$; then

$$\cos \{[(p+1)/n_1]\pi\} = \sin (\pi/2n_1) \quad [13a]$$

so Equation [11] becomes

$$\psi = \arcsin (\cos \lambda \sin [\pi/2n_1]) \quad [14a]$$

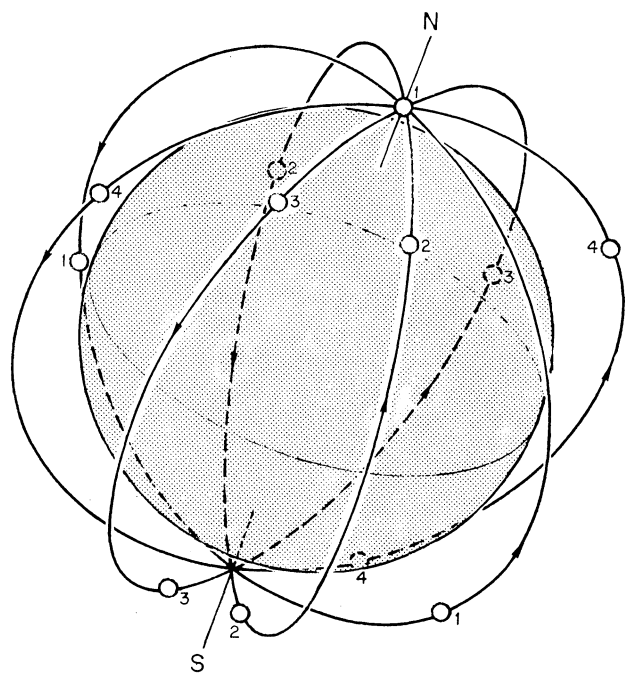
However, if n_1 is even, p is equal to $(n_1 - 4)/2$; then

$$\cos \{[(p+1)/n_1]\pi\} = \sin (\pi/n_1) \quad [13b]$$

in which case Equation [11] becomes

$$\psi = \arcsin (\cos \lambda \sin [\pi/n_1]) \quad [14b]$$

The discrepancy between Equations [14a and 14b] was anticipated earlier in the discussion, when it was realized that,



EXAMPLE:

$$\begin{aligned} n_1 &= 4 \\ n_2 &= 3 \end{aligned}$$

ASSUMPTIONS

1. ALL ORBITS ARE CIRCLES OF COMMON SIZE.
2. THE PLANES ARE EVENLY FANNED ABOUT THEIR COMMON AXIS, NS.
3. THE SATELLITES ARE EVENLY SPACED IN ORBIT.

Fig. 7 Diagram of the polar satellite network

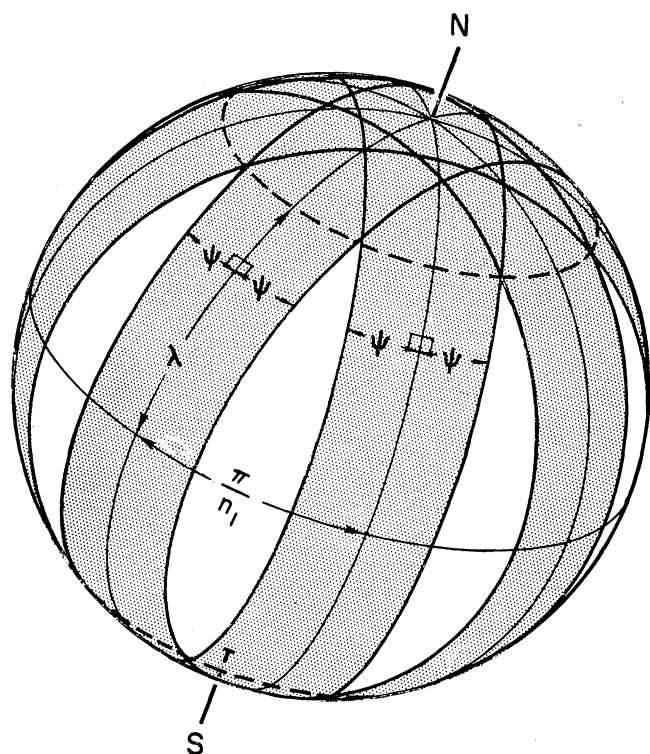


Fig. 8 Coverage geometry for a polar satellite network

for instance, a four-plane inclined network degenerated at $i = 90$ deg to a double two-plane polar network. However, meaningful results may be obtained for a polar network containing any (even or odd) number of distinct orbit planes. To find ψ required for a two-plane polar network, one could set n_1 equal to 4 in Equation [14b] to obtain ψ and substitute that value of ψ and n_1 in Equation [4] to obtain N , the number of satellites actually required in a double two-plane network. But only half that number of satellites is required in a polar network consisting of two *distinct* orbit planes. Essentially then, whether n_1 is odd or even, Equation [14a] can be applied to determine ψ . Consequently, Equation [11] becomes

$$\psi = \arcsin [\cos \lambda \sin (\pi / 2 n_1)] \quad [15]$$

Fig. 8 illustrates the manner in which coverage of the polar caps down to latitude λ is effected by overlapping zones of coverage swept out by the satellites in neighboring orbit planes.

Equations [4 and 15] were solved to obtain N , the total number of satellites required, as a function of system altitude h and the number of orbit planes n_1 for the case of complete, global coverage ($\lambda = 0$ deg) with ideal sensors (i.e., $\sigma = 0$ deg). The results are displayed in Fig. 9. The rather sharp heels appearing in the curves of constant n_1 indicate that it is undesirable (if satellite altitude is to be minimized) to pack an excessively large number of satellites into each plane. Instead it is better to increase n_1 the number of orbit planes, and to decrease n_2 the satellite density per plane, in order to reduce the system altitude at some N level. For example, 36 satellites may be arranged as follows to provide complete global coverage:

Orbit planes	Satellites/plane	Required altitude
$n_1 = 2$	$n_2 = 18$	1500 nautical miles
3	12	672
4	9	522
6	6	672
9	4	1500

Then, from a minimum altitude standpoint, a network of 4×9 is most desirable for complete global coverage if a polar network consisting of 36 satellites is to be employed.

Fig. 10 presents the envelope to the n_1 family of curves shown in Fig. 9. The requirement that both n_1 and n_2 be integers results in the step-like practical optimum curve. The variations in step size and frequency result from the irregular occurrence of N values which may be factored in an optimum or near-optimum fashion. For example, as was seen previously, the minimum altitude of a 36-satellite polar network which provides complete global coverage is 522 nautical miles. In order to reduce the network altitude further, N must be increased to at least 40 ($h = 475$ nautical miles), since the best

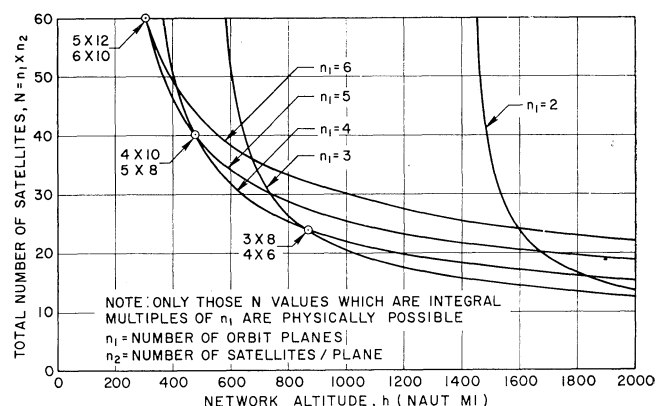


Fig. 9 Number and distribution of satellites in a polar network providing complete global coverage ($\lambda = 0$ deg, $\sigma = 0$ deg)

38- and 39-satellite networks require altitudes of 1492 and 651 nautical miles, respectively. Note that when N is twice the square of an integer ($2n_1^2$), the theoretical optimum may actually be attained.

The theoretical optimum curves for $\sigma = 10$ and 20 deg demonstrate the strong dependence of N on σ . The practical optimum curves for each of these σ values may be obtained by a point-by-point transformation of the practical optimum curve for $\sigma = 0$ deg by means of the curves in Fig. 2.

The theoretical optimum satellite utilization curves for several nonzero values of λ were obtained in a similar manner and are presented in Fig. 11. Note that sizable reductions in system requirements may be effected by modification of the coverage requirement. For example, a system of 24 satellites providing complete global coverage requires an altitude of 859 nautical miles; but if $\lambda = 30$ deg, the altitude need be only 691 nautical miles.

Equations [3, 4 and 15] indicate the following dependence of system altitude upon the number of orbit planes in a polar network

$$h = R_E \left\{ \frac{\cos \sigma}{\cos (\arccos \{ \cos (\pi/n_2) \cos [\arcsin (\cos \lambda \sin \pi/2n_1)] \} + \sigma)} - 1 \right\} \quad [16]$$

Minimization of h with respect to n_1 , holding N , σ and λ constant, results in the expression

$$\frac{4n_1}{n_2} \frac{\tan \pi/n_2}{\cos^2 \lambda \sin \pi/n_1} \left(1 - \cos^2 \lambda \sin^2 \frac{\pi}{2n_1} \right) - 1 = 0 \quad [17]$$

For the special case of complete global coverage mentioned earlier, λ is equal to zero deg and Equation [17] takes the form

$$2n_1 \tan \pi/n_2 = n_2 \tan \pi/2n_1 \quad [18]$$

whose solution is simply

$$n_2 = 2n_1 \quad [19]$$

In practice, an integral value of N will not generally be twice the square of an integer ($2n_1^2$) permitting such a theoretical optimum distribution. A convenient rule of thumb for selecting the practical network which minimizes system altitude is to choose the pair of integer factors of N which minimizes quantity q defined by

$$q = |n_2 - 2n_1| \quad [20]$$

It is clear from the manner in which n_1 and n_2 are involved in Equation [16] why the (3×10) and the (5×6) polar networks mentioned earlier require identical altitudes to provide complete global coverage.

More generally, it appears that the minimum-altitude polar network of N satellites providing coverage down to latitude λ is the one for which q_λ assumes its minimum value, where

$$q_\lambda = |n_2 - 2n_1/\cos \lambda| \quad [21]$$

Case B: The Equatorial Network

In this case i is equal to zero deg, and a single orbit plane is used to avoid redundancy. From Equation [5] it is clear that

$$\psi = \mu \quad [22]$$

That is, the zone from $-\mu$ to $+\mu$ will be continuously observed by the equatorial chain of satellites. Then from Equations [4 and 22] it follows that

$$h = R_E \{ \cos \sigma / \cos [\arcsin (\cos \pi/N \cos \mu) + \sigma] - 1 \} \quad [23]$$

Equation [23] was solved to obtain curves of N vs. μ and h for a network consisting of a single equatorial orbit plane. The results are displayed in Fig. 13 together with the system requirements for multiplane inclined networks examined in case C. Discussion of the results is deferred until that time.

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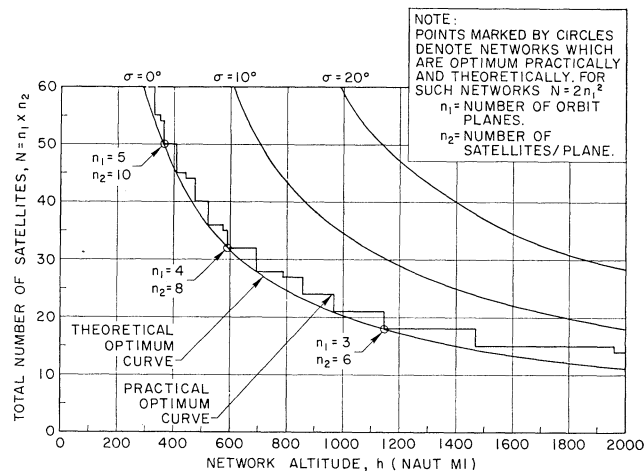


Fig. 10 Theoretical and practical optimum satellite utilization in a polar network providing complete global coverage ($\lambda = 0$ deg)

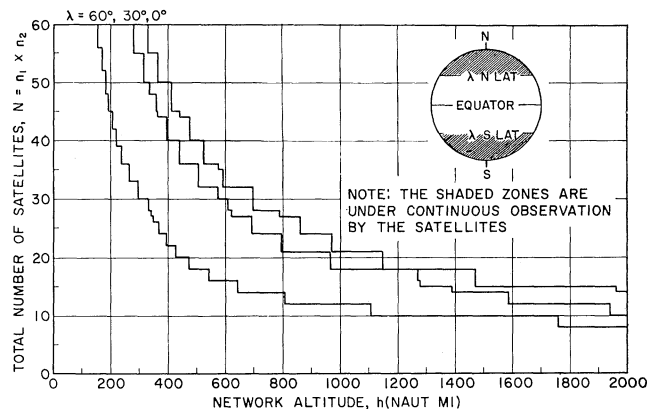


Fig. 11 Comparison of polar satellite networks providing coverage from the poles to minimum latitude λ . ($\sigma = 0$ deg)

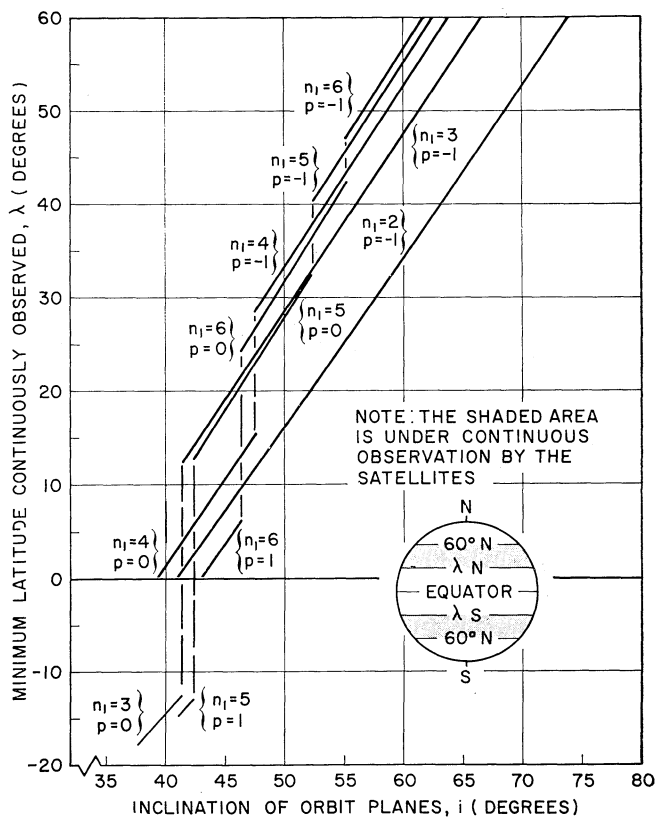


Fig. 12 Minimum latitude continuously observed as a function of inclination and number of orbit planes in network ($\mu = 60$ deg)

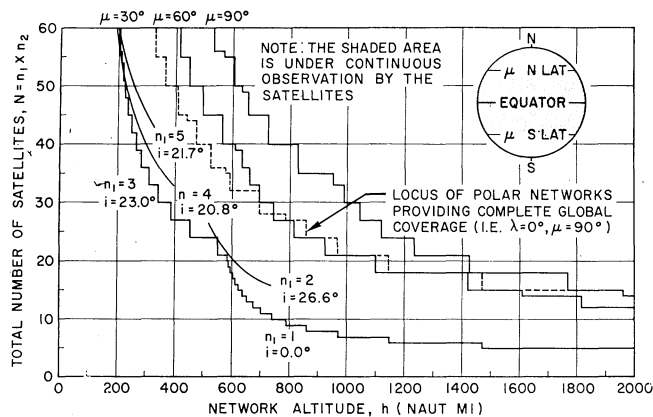


Fig. 13 Comparison of inclined and polar satellite networks providing coverage from the Equator to maximum latitude μ . ($\sigma = 0$ deg)

Case C: The General Network

In the most common case, the orbit inclination will not necessarily be zero or 90 deg, nor will λ or μ assume its extreme value of zero or 90 deg. For example, suppose coverage is required from a minimum latitude λ to a maximum latitude μ of 60 deg. Substitution of values of μ , n_1 and i in relations [5-7] yields λ as a function of those parameters as shown in Fig. 12.

Because of the role of the integer p in Equation [5], λ is a piecewise continuous function of i for $n_1 > 2$. The physical explanation for the discontinuous nature of λ is as follows. As the orbit inclination i decreases, ψ must increase so as to maintain coverage up to latitude μ . Then, when the northernmost mesh (below latitude i) becomes covered, point V (the southernmost continuously observed point) will suddenly appear in the next mesh to the south. As ψ increases further, λ will decrease continuously until that mesh is covered—and so on. For odd values of n_1 , λ will assume negative values until ψ reaches the value $\arccos \sqrt{1 - \sin^2 i \sin^2 (\pi/n_1)}$, when the swaths covered from adjacent ascending satellite orbits overlap everywhere. Then λ becomes undefined with coverage extending from $-\mu$ to $+\mu$.

It is clear from Fig. 12 that the appropriate inclinations of networks providing coverage from $\lambda = 0$ deg to $\mu = 60$ deg are

$i = 40.9$ deg	for	$n_1 = 2$
41.4 deg		3
39.2 deg		4
42.4 deg		5

Substitution of these (n_1, i) pairs in Equations [4 and 6] yields N as a function of h and n_1 for $\lambda = 0$ deg and $\mu = 60$ deg. Similarly, curves may be obtained for other values of λ and μ . Fig. 13 shows the results appropriate to networks providing coverage from the Equator to latitude μ in each hemisphere. The N vs. h curve for the polar network was included in the figure as a basis for comparison of the two kinds of networks for various coverage requirements.

Fig. 12 indicates that, in order to cover the latitude belt from $-\mu$ to $+\mu$, a network consisting of an even number of orbit planes must observe all points down to latitude $\lambda = 0$ deg. However, a network in which n_1 is odd need not observe all points down to the Equator in order to cover the zone from $-\mu$ to $+\mu$; that is, for n_1 odd, the most remote point to be observed will be north of the Equator. Note that in Fig. 13 for $\mu = 30$ deg, networks for which n_1 is odd are superior to those for which n_1 is even; above a system altitude of 582 nautical miles, a single equatorial orbit plane should be used to provide the required coverage, but immediately below that altitude a three-plane network is preferred.

It appears from Fig. 13 that at low system altitudes a polar network providing complete global coverage requires fewer satellites than does an inclined network providing coverage of a wide equatorial zone. The altitude at which the polar and the inclined networks require equal numbers of satellites increases with μ . For $\mu = 90$ deg, the inclined and polar networks require equal numbers at an altitude of 1960 nautical miles, above which the inclined network consists of two orbit planes. Since each of the planes is inclined at 45 deg with respect to the equatorial plane, the polar and inclined networks are geometrically identical for a 90-deg rotation of the axis common to the orbit planes.

Conclusions

A method was developed to determine the minimum number of satellites required to provide continuous coverage of a latitudinally bounded zone by means of a particular kind of satellite network.

Several representative cases of coverage requirements were examined with the following results:

1 The polar network is preferable to the inclined network in providing complete global coverage. Ideally, the distribution of satellites in it is such that the number of satellites in each orbit plane is twice the number of orbit planes.

2 The number and inclination of the orbit planes in the optimum inclined network designed to satisfy a particular coverage requirement vary discontinuously with system altitude.

3 To determine the properties of the optimum satellite network satisfying a certain coverage requirement, it is *generally* necessary to compare the N vs. h performance curves obtained separately for the polar, equatorial and inclined networks. From such performance curves, preferred orbital inclinations and distributions can be chosen for satellite networks at pre-assigned altitudes.

Acknowledgment

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Nomenclature

h	= altitude of each satellite in the network
i	= inclination of each orbit plane to the Equator
n_1	= number of orbit planes in the network
n_2	= number of satellites in each orbit plane
N	= total number of satellites in the network
R_E	= radius of Earth (= 3440 nautical miles) or other central body
S	= slant range (Fig. 2)
θ	= central angle subtended by the observed great circle half-arc (Fig. 1)
λ	= minimum latitude to be observed continuously
μ	= maximum latitude to be observed continuously
σ	= minimum angle of visibility (Fig. 1)
ψ	= great circle arc half-width of observed swath

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